



FATIMA JINNAH WOMEN UNIVERSITY

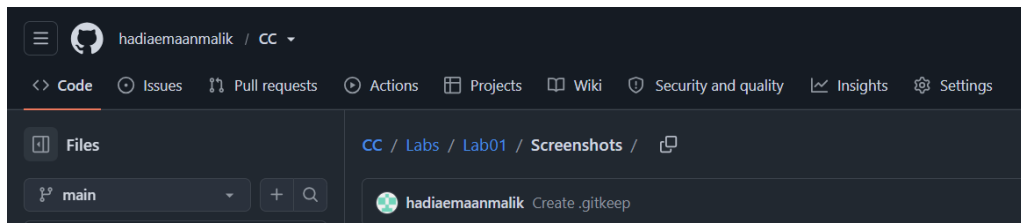
Department of Software Engineering

Lab 01

“ GitHub Identity, Student Portal Registration, and Ubuntu Server Installation ”

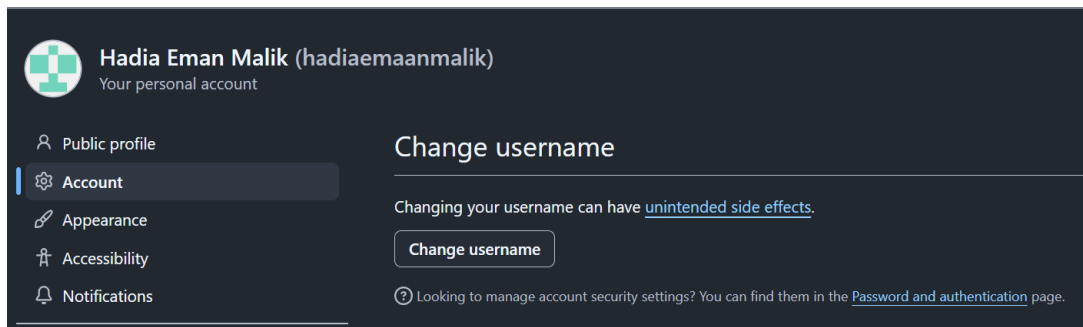
Name	:	Hadia Eman Malik
Registration No.	:	FA24B1-SE-025
Course Code	:	BSE 665
Section	:	A
GitHub Username	:	hadiaemaanmalik

Getting Started:

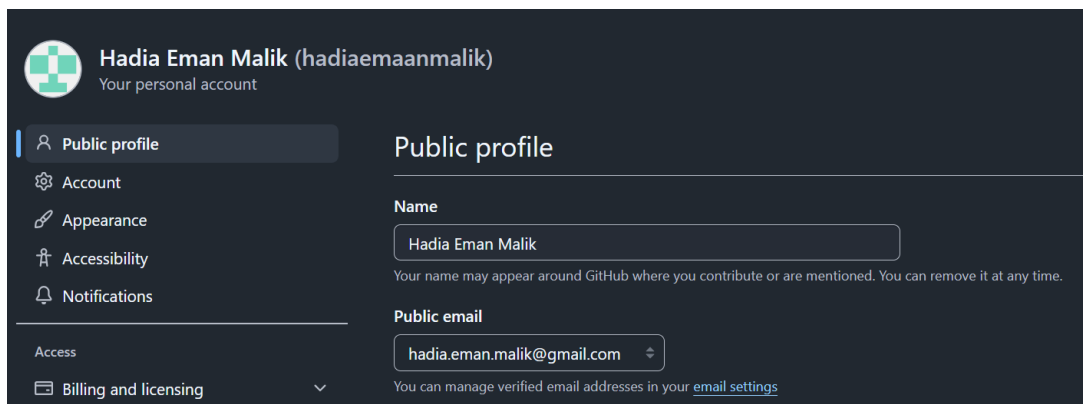


TASK 1

- **Create and Configure Your GitHub Account**



The GitHub sign-up form with fields for username, email, and password being filled in to create a new account.



GitHub profile dashboard showing the username 'hadiaemaanmalik'

TASK 2

- **Register in the Student Marks Portal**



Register

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

Section

Submit enrollment

Sign out

Student Marks Portal registration page where personal and academic details are entered

Enrollment Pending

STUDENT MARKS PORTAL

Course enrollments

Sign out



Hadia Eman Malik
2024-BSE-25

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section A

Waiting for administrator approval

Pending approval

"Pending" enrollment status, indicating that the registration has been received but not yet processed

Enrollment Submitted

STUDENT MARKS PORTAL

Academic Record

Sign out



Hadia Eman Malik
GitHub authenticated

Registration
2024-BSE-25

GitHub username
hadiaemaanmalik

Email
hadia.eman.malik@gmail.com

Approved courses
1

+ Enroll in another course

APPROVED COURSES

Course Marks

Not published

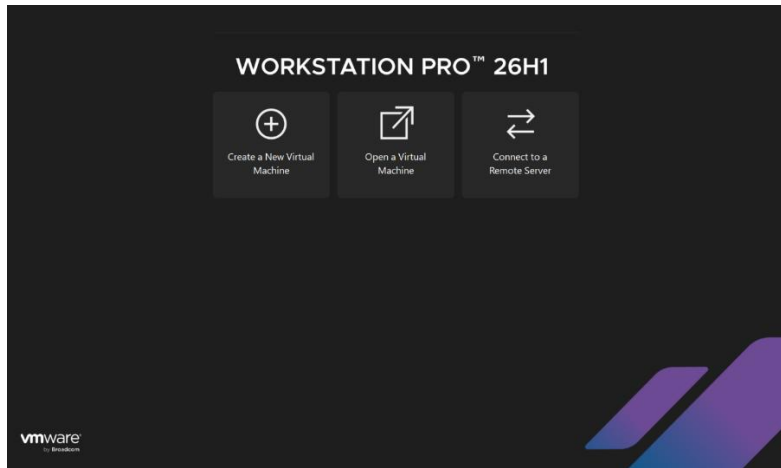
Cloud Computing (SE-665) Section A

ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Marks have not been published.			
Total	—	—	—

Confirmation screen showing that the enrollment form has been successfully submitted for review.

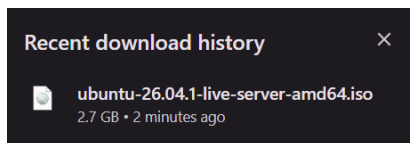
TASK 3

- **Download VMware Workstation Pro and Ubuntu Server**



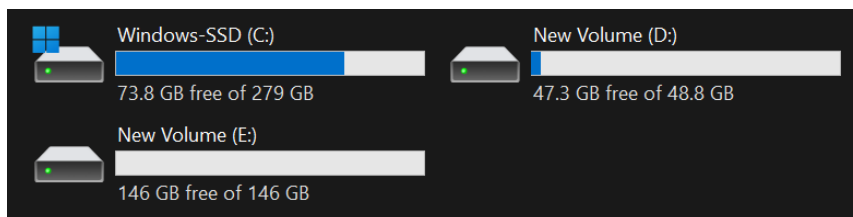
VMware download page where VMware Workstation Pro is being downloaded for use as the virtualization platform

Ubuntu ISO File



The Ubuntu Server ISO image being downloaded from the official Ubuntu releases page, ready for installation in the virtual machine

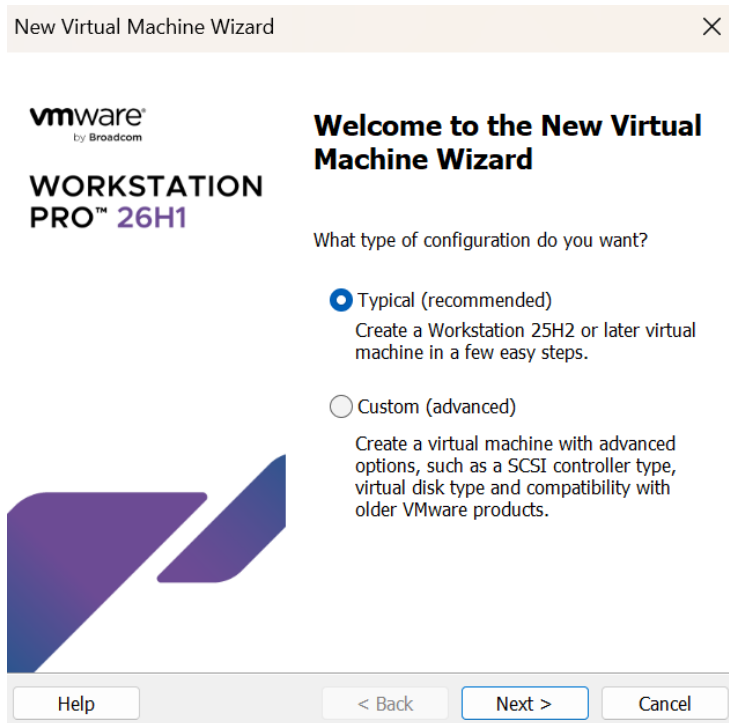
Available Storage



Stored with sufficient free disk space available

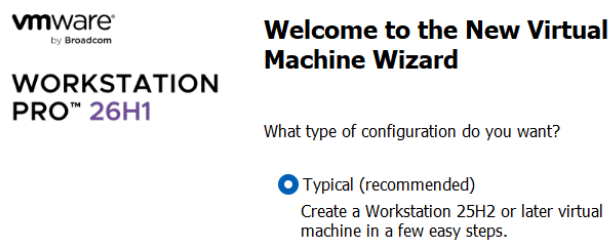
TASK 4

- **Create the Ubuntu Server Virtual Machine**



VMware "New Virtual Machine Wizard" with "Typical (recommended)" selected as the configuration type

Select Typical



Browse Ubuntu Server File

New Virtual Machine Wizard ✕

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:
No drives available

☒ Installer disc image file (iso):

⇒ Select the installer disc image to continue.

☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

The file browser dialog in VMware pointing to the downloaded Ubuntu Server ISO file, which will serve as the installation medium for the virtual machine

New Virtual Machine Wizard ✕

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:
No drives available

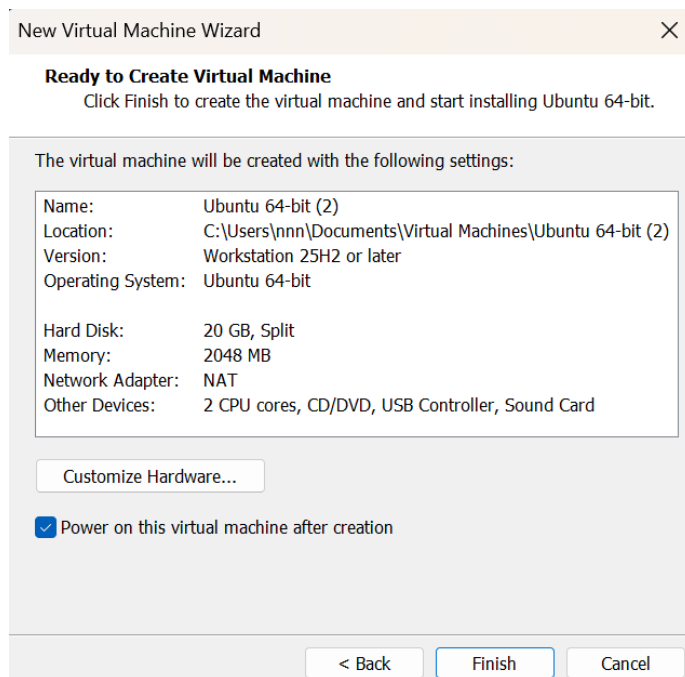
☒ Installer disc image file (iso):

ℹ Ubuntu 64-bit Server 26.04.1 detected.
To use Easy Install, insert the first disc of the set.

☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

The wizard screen where the virtual machine is given a descriptive name and a storage location on the host machine is specified

Configuration Summary



A summary screen listing all the chosen VM settings — name, location, disk size, and memory.

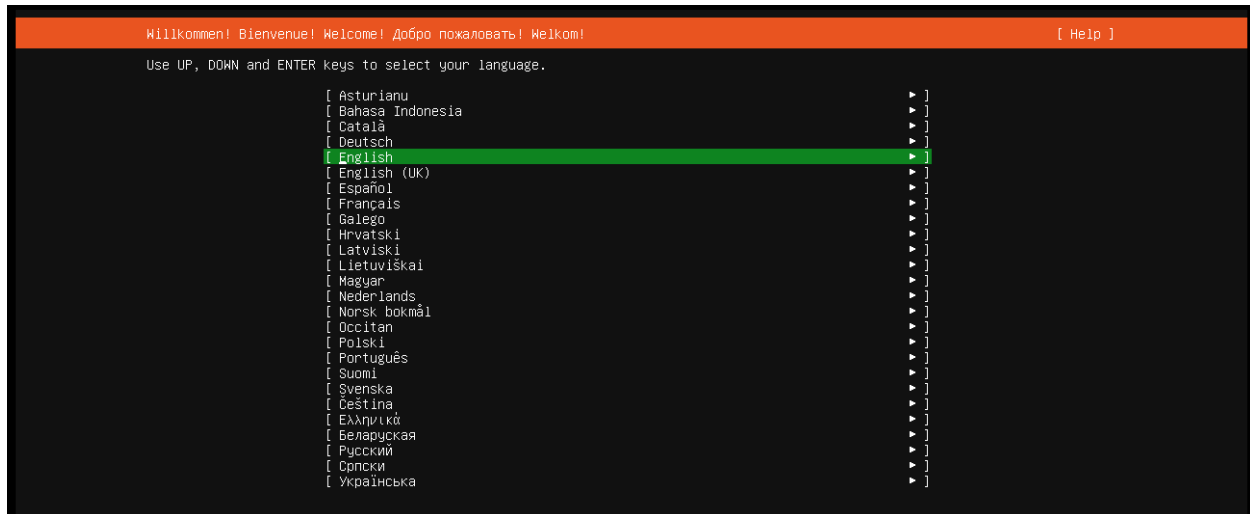
Ubuntu Installer Boot Waiting



The Ubuntu Server installer loading screen with progress indicators, showing that the kernel and initial setup modules are being loaded into memory.

TASK 5

- **Choose the Installation Language**

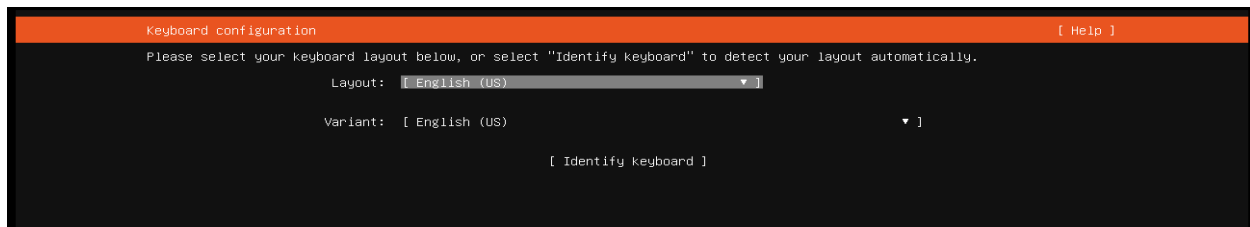


The Ubuntu Server installer language selection menu with "English" highlighted, which determines the language used throughout the installation and the final system.

TASK 6

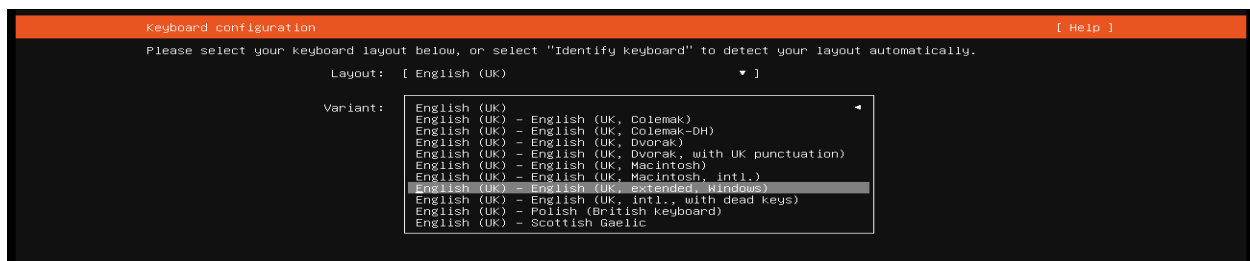
- **Configure the Keyboard Layout**

Layout



The installer keyboard layout list with the appropriate layout, 'English (UK)' selected to match the physical keyboard being used.

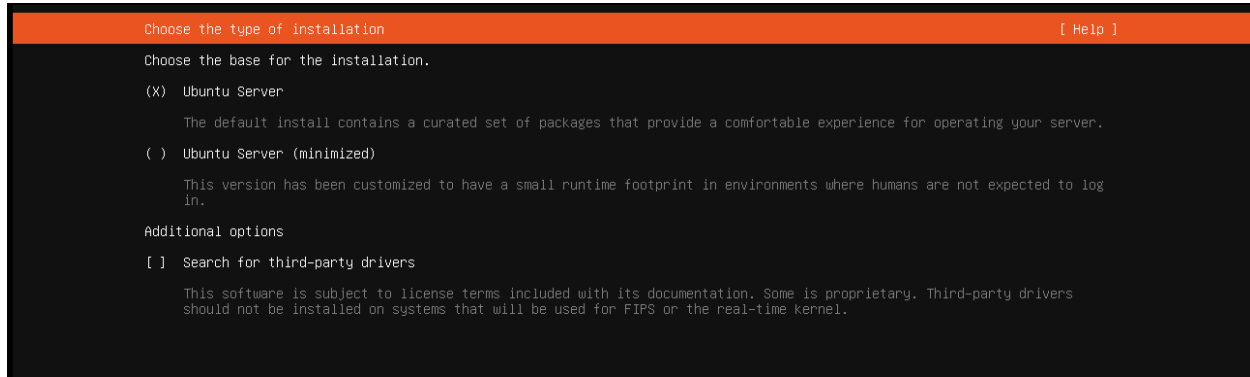
Variant



The keyboard variant sub-menu confirming the specific variant of the chosen layout, ensuring correct key mapping for all characters and symbols.

TASK 7

- **Choose the Ubuntu Server Installation Type**

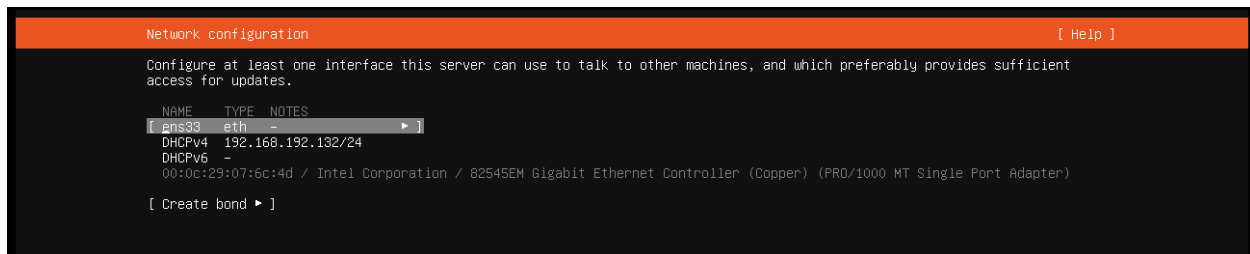


The installer screen presenting installation options, with Ubuntu Server selected for a full-featured server setup.

TASK 8

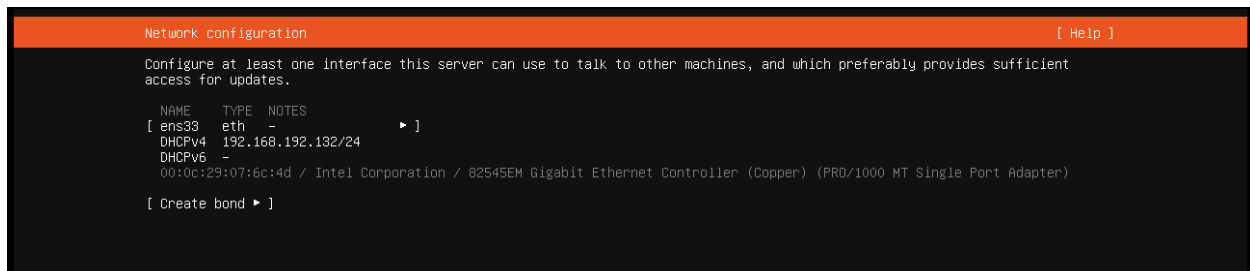
- **Configure the Network**

Network Interface



The installer displaying the detected network interface (typically ens33 or eth0), confirming that the virtual machine's virtual NIC has been recognized.

Installer Network

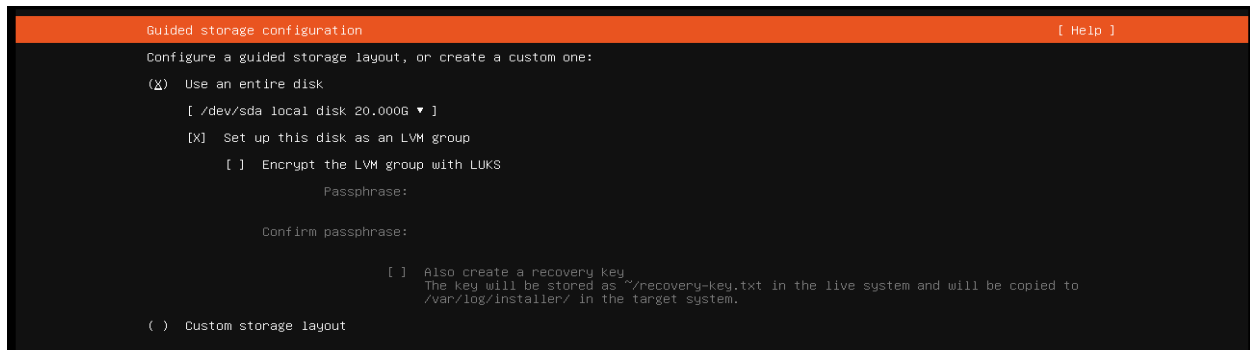


The network configuration screen showing the dynamically assigned IPv4 address obtained through DHCP, establishing the server's network connectivity.

TASK 9

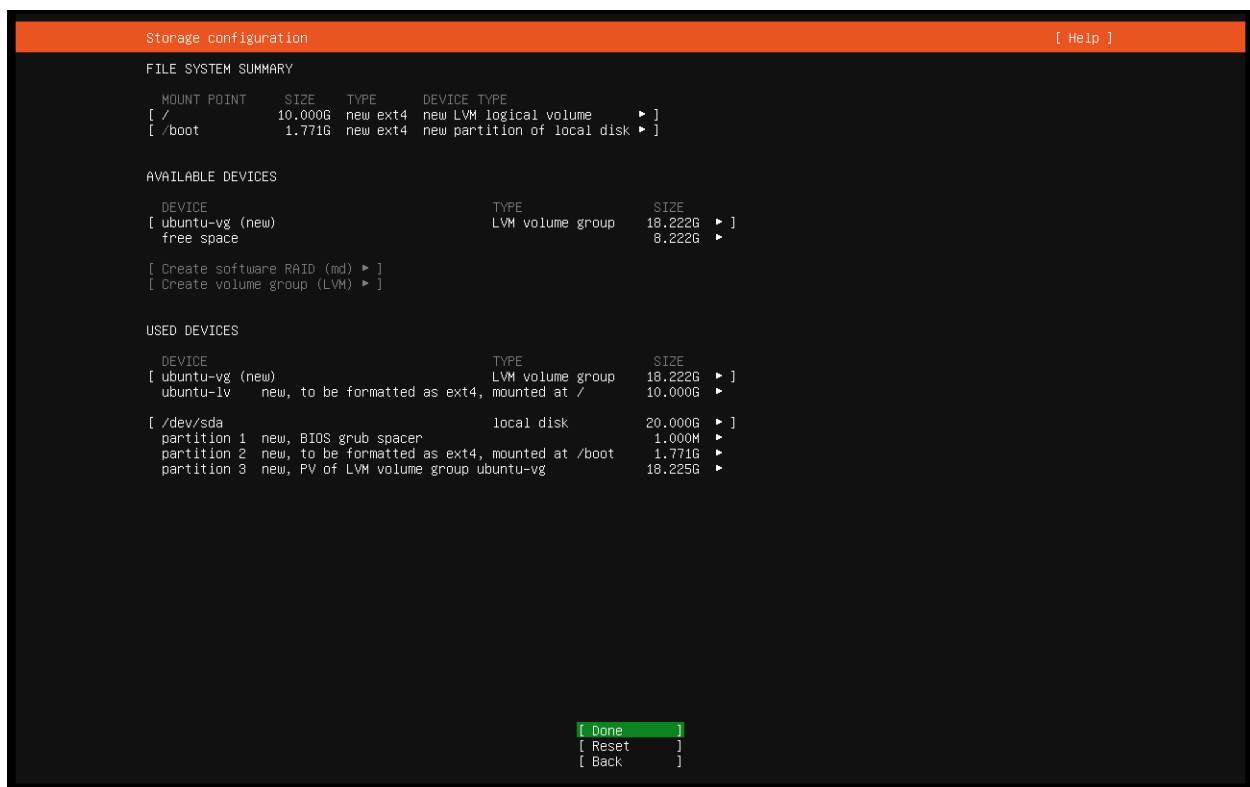
- **Configure Storage**

Guided Storage



The storage configuration screen with "Use an entire disk" selected as the guided storage method, which will automatically partition the virtual disk.

Storage Configuration

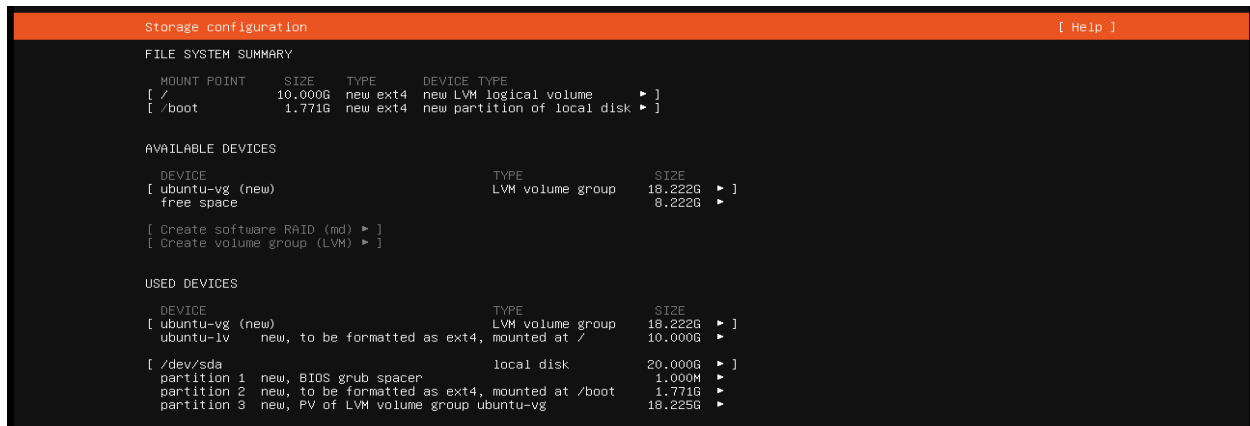


The summary of the proposed storage layout showing the disk partition plan, including the root filesystem and any additional partitions before any changes are applied.

TASK 10

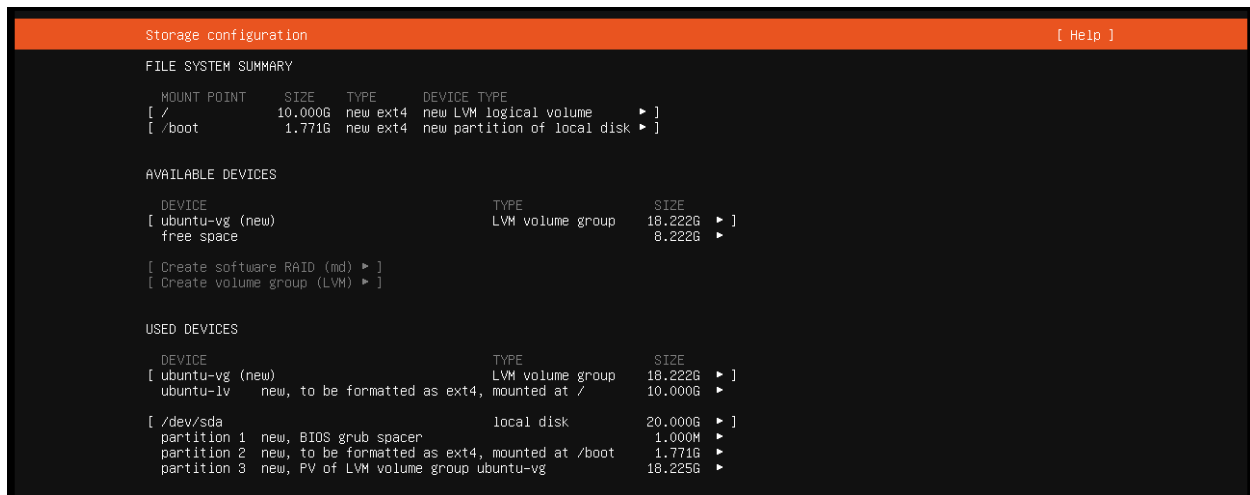
- **Select the Installation Disk**

Selected Disk



The installer disk selection screen with the available virtual disk highlighted, typically /dev/sda, chosen as the target for Ubuntu Server installation.

Disk Detail



A detailed view of the selected disk showing its total capacity, model identifier, and confirming that it is the correct destination for the operating system.

TASK 11

- Review the Partitions

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /            10.000G new ext4 new LVM logical volume ▶ ]
[ /boot        1.771G  new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
free space      8.222G ▶

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶

[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶ ]
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
partition 3     new, PV of LVM volume group ubuntu-vg 18.225G ▶ ]
```

The full partition layout presented for review, listing each partition, its mount point, size, and filesystem type before the installer writes the changes to disk.

TASK 12

- Confirm the Storage Changes

Storage Warning

```
Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

Are you sure you want to continue?

[ No ]
[ Continue ]
```

A warning prompt alerting that proceeding will destroy all existing data on the selected disk and that the action is irreversible

Storage Confirmed

```
Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

Are you sure you want to continue?

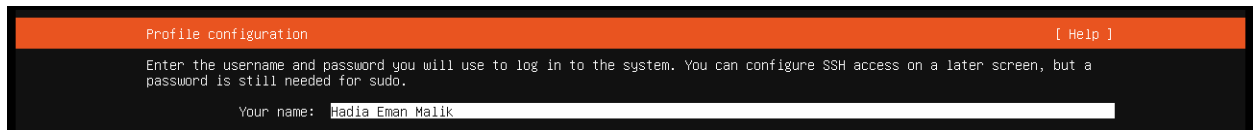
[ No ]
[ Continue ]
```

The "Confirm destructive action" dialog with "Continue" selected, giving final approval for the installer to format the disk and create the new partition table.

TASK 13

- **Configure the Ubuntu Profile**

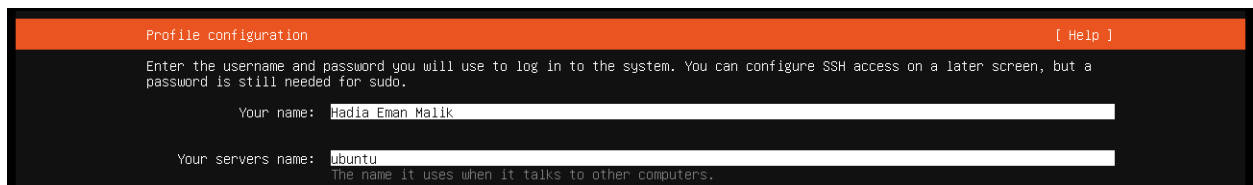
Actual Name



The screenshot shows the 'Profile configuration' window with an orange header bar containing '[Help]'. Below the header, text reads: 'Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.' The 'Your name:' label is followed by a text input field containing 'Hadia Eman Malik'.

The profile configuration screen with the user's full name entered.

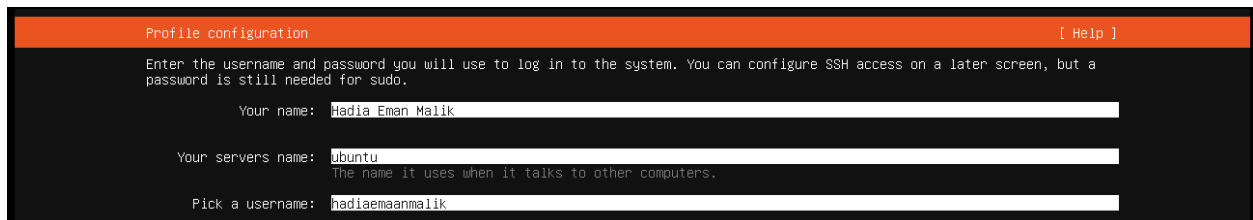
Server Name



This screenshot shows the 'Profile configuration' window with the 'Your name' field filled with 'Hadia Eman Malik'. The 'Your servers name:' label is followed by a text input field containing 'ubuntu'. Below this field, smaller text reads: 'The name it uses when it talks to other computers.'

The "Your server's name" field filled in with a hostname, typically a short, recognizable label used to identify the machine on the network.

Username



This screenshot shows the 'Profile configuration' window with the 'Your name' and 'Your servers name' fields filled. The 'Pick a username:' label is followed by a text input field containing 'hadiaemaanmalik'.

The "Pick a username" field showing the chosen system username, which will be used for authentication and as the home directory name on the server.

Review Profile

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

A final review of the profile details, real name, server name, and username, displayed together before the installer proceeds to copy system files.

TASK 14

- Complete the Software Installation

Updating system

[Help]

```
configuring partition: partition-2
configuring format: format-0
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
  running 'curtin extract'
  curtin command extract
  acquiring and extracting image from cp:///tmp/tmpq520qvg6/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
```

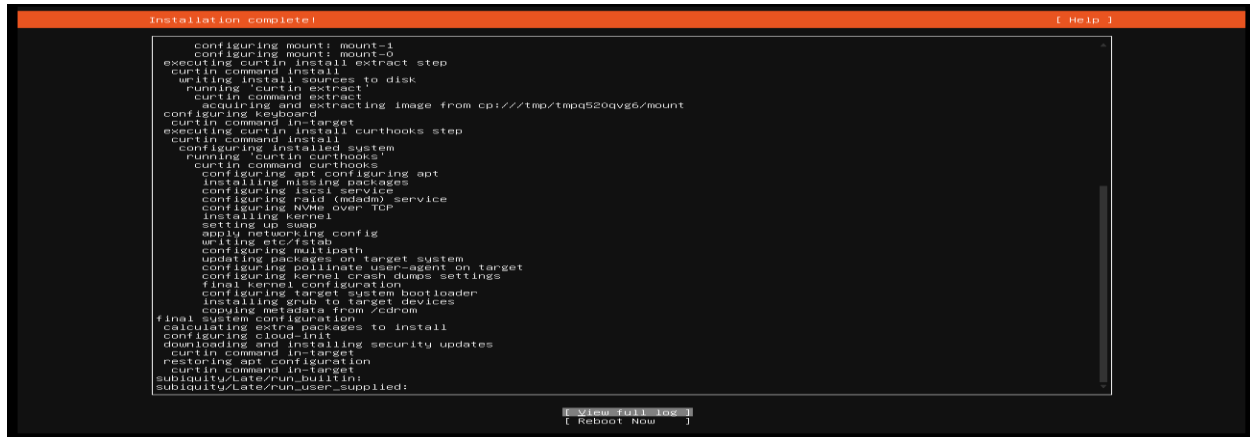
[View full log]

The installer's progress bar showing that system files are being copied, packages are being extracted, and the base Ubuntu Server system is being assembled on the target disk.

TASK 15

- **Reboot and Verify the Ubuntu Identity**

Installation Complete



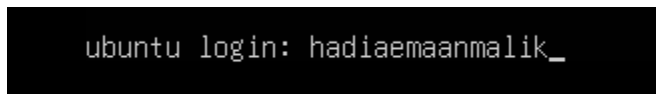
```
Installation complete! [ Help ]

configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/q520qvsg6/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring nvme over TCP
installing kernel
setting up Sshd
apply networking config
writing etc/rsync
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from zcdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run-built-in
subiquity/late/run-user-supplied:

[ View full log ]
[ Reboot Now ]
```

The post-installation screen confirming that Ubuntu Server has been installed successfully and prompting the user to remove the installation medium and reboot.

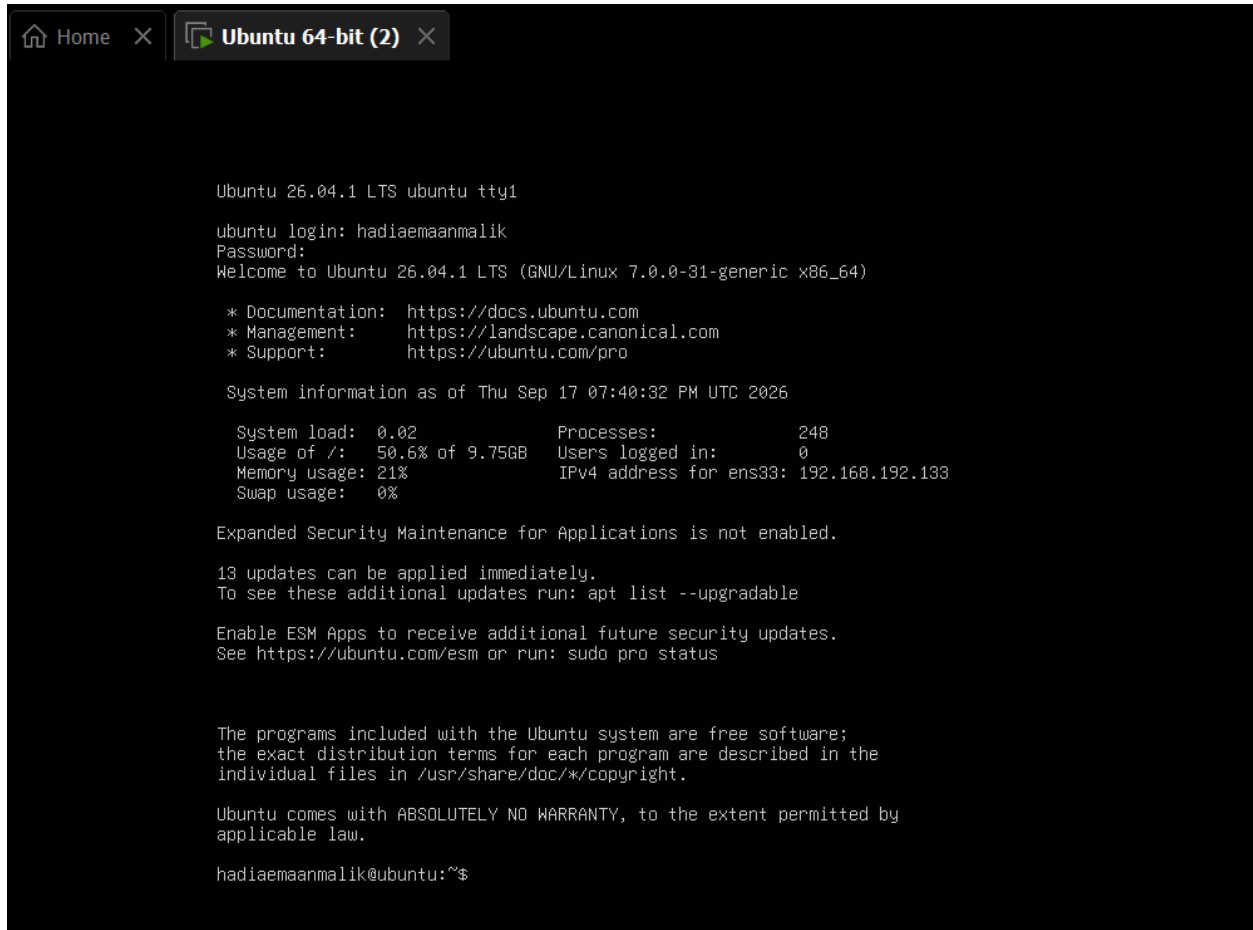
Ubuntu Login Screen



```
ubuntu login: hadiaemaanmalik_
```

The text-based login prompt appearing after the system reboots, with the server hostname displayed and awaiting the username and password credentials.

Ubuntu Terminal Screen

A screenshot of an Ubuntu terminal window. The window has a title bar with a 'Home' button and a tab labeled 'Ubuntu 64-bit (2)'. The terminal output shows the login process for user 'hadiaemaanmalik' on Ubuntu 26.04.1 LTS. It displays system information, including system load, memory usage, and available updates. The prompt is 'hadiaemaanmalik@ubuntu:~\$'.

The command-line shell session immediately after a successful login, displaying the welcome banner, system information summary, and a ready Bash prompt.

Ubuntu Identity Verified

A terminal snippet showing the command 'whoami' being executed, with the output 'hadiaemaanmalik@ubuntu:~\$ _'.

command, which returns the logged-in username, confirming that the correct user identity is active on the server.

TASK 16

- **Find the Ubuntu Server IP Address**

IP addr Command

```
hadiaemaanmalik@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:07:6c:4d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29076c4d
    inet 192.168.192.133/24 metric 100 brd 192.168.192.255 scope global dynamic ens33
        valid_lft 1115sec preferred_lft 1115sec
    inet6 fe80::20c:29ff:fe07:6c4d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hadiaemaanmalik@ubuntu:~$
```

The ip addr command being executed in the terminal, which lists all network interfaces and their assigned IP addresses in detail.

Ubuntu IP Address

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:07:6c:4d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29076c4d
    inet 192.168.192.133/24 metric 100 brd 192.168.192.255 scope global dynamic ens33
        valid_lft 1115sec preferred_lft 1115sec
    inet6 fe80::20c:29ff:fe07:6c4d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hadiaemaanmalik@ubuntu:~$
```

The command output showing the server's IPv4 address (typically a 192.168.x.x address on the ens33 or eth0 interface), ready to be used for remote SSH connections.

TASK 17

- **Connect to Ubuntu Server from Windows**

Windows ssh Command

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nnn>ssh hadiaemaanmalik@192.168.192.133
The authenticity of host '192.168.192.133 (192.168.192.133)' can't be established.
ED25519 key fingerprint is SHA256:GoMS4a/tHd0PwDojPZnBuAc3hUULsiJTgyTIseV0IaI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

The Windows command prompt or PowerShell window with the ssh username@ip-address command typed, initiating a secure remote connection to the Ubuntu Server.

ssh Fingerprint

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.192.133' (ED25519) to the list of known hosts.
ssh_dispatch_run_fatal: Connection to 192.168.192.133 port 22: Connection aborted

C:\Users\nnn>ssh hadiaemaanmalik@192.168.192.133
hadiaemaanmalik@192.168.192.133's password: |
```

The SSH "authenticity of host" warning displayed on first connection, showing the ECDSA key fingerprint and asking whether to trust and continue connecting.

ssh Login

```
C:\Users\nnn>ssh hadiaemaanmalik@192.168.192.133
hadiaemaanmalik@192.168.192.133's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 08:14:02 PM UTC 2026

System load:  0.0               Processes:           219
Usage of /:   50.8% of 9.75GB   Users logged in:    0
Memory usage: 24%              IPv4 address for ens33: 192.168.192.133
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

hadiaemaanmalik@ubuntu:~$ |
```

The SSH session prompting for the Ubuntu user's password after the host key has been accepted, authenticating the remote login request.

ssh Identity

```
hadiaemaanmalik@ubuntu:~$ |
```

A fully established SSH session from Windows into the Ubuntu Server, with the shell prompt visible, confirming remote access and verifying the same user identity as seen locally.